NPHI101: Engineering Physics
Mid-Semester Examination 2025-26 (Monsoon Semester)

Total Marks: 30 Time: 1 hour

Use separate answer sheets for different parts. Mention the Part No. at the top.   **All the symbols have their usual meaning.**

PART 1 (CLASSICAL MECHANICS AND ELECTRODYNAMICS)

1. A bead of mass m slides without friction on a frictionless wire in the shape of a cycloid, with equations, $x = a(\theta - \sin\theta)$, $y = a(1+\cos\theta)$, $0 \leq \theta \leq 2\pi$, and a is a constant. Find (a) the Lagrangian function L (in terms of θ, $\dot{\theta}$), (b) the Euler-Lagrange equation of motion. Find (c) an expression for the generalized momentum p_θ, and (d) the Hamiltonian of the system in terms of p_θ and L.

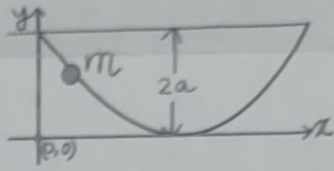

Figure: Bead sliding on a cycloid

Marks: 2+2+1+2=7

2. Show that the force field given by, $\vec{F} = x^2yz\hat{i} - xyz^2\hat{k}$, is non-conservative. **Marks: 3**

PART 2 (THERMAL AND STATISTICAL PHYSICS)

1. (a) Consider a gas that obeys the modified equation of state: $P(V-b) = Nk_BT$, where b is a constant, N is the number of particles, k_B is Boltzmann's constant. The gas is expanded isothermally where the volume increased from V_0 to $4V_0$. Find the work done (in terms of b, N, k_B, V_0 and T) by the gas. Sketch the P-V diagram for this process indicating the initial and final states and shade the region that represents the work done during the expansion. **Marks: 2+2**

   (b) The Helmholtz free energy of N number of molecules is given by $F = N\epsilon_0 - N\beta V - N\alpha T\ln(T) - Nk_BT\ln(V/N)$. Here ϵ_0, β, and α are constants. Calculate the pressure of the system. **Marks: 2**

2. The radiation emitted by cavity (black body) oscillators has discrete energy values: 0, hc/λ, $2hc/\lambda$, $3hc/\lambda$, and so on (here λ is the wavelength). Calculate the average energy per oscillator. **Marks: 4**

PART 3 (MODERN PHYSICS)

1. The lifetimes associated with the atomic spontaneous decays from E_3 to E_2 and from E_3 to E_1 are 70 ns and 200 ns, respectively. Calculate the lifetime of E_3 for the overall spontaneous decay from that level? **Marks: 3**

2. Explain the pumping model for a four-level laser and derive an expression for a steady-state population inversion between the lasing levels. **Marks: 3**

3. From the probabilistic interpretation, show why the wavefunction Ψ of a quantum particle should be square-integrable. **Marks: 2**

4. Assume that $\Psi(x) = Ax^4$, where A is a suitable constant. What kind of quantum particle does it represent? Explain. **Marks: 2**

$$\frac{dP}{dV} = -\frac{Nk_BT}{(V-b)^2}$$